Tab 1

EUS liver analysis combines **detection, tracking, temporal reasoning, and classification**. I'll structure the explanation so you can treat it like a **mini research roadmap**.

---

1. First: What exactly must be predicted from the CEUS video?

Before designing architectures, we must define the **outputs** the system needs.

From a **2-minute CEUS video**, the system should ideally extract:

A. Lesion localization (spatial)

For each frame:

- bounding box / segmentation mask of lesion
- lesion centroid
- lesion size (diameter/area)
- lesion shape features

This is needed because **lesions move due to breathing and probe motion**.

B. Lesion tracking (temporal)

Once detected:

- track the **same lesion across frames**
- estimate motion trajectory

Outputs:

- lesion ID

- time-indexed bounding boxes

Example:

Frame	Lesion X	Lesion Y	Area
1	120	200	340
2	123	205	336

C. Enhancement pattern over time

CEUS uses **contrast microbubbles**.

Key patterns:

Phase	Typical Time
Arterial	~10–30 sec
Portal	~30–120 sec
Late	>120 sec

For each lesion we must measure:

- intensity change
- enhancement strength
- wash-in speed
- wash-out speed

So we extract:

Enhancement curve = intensity(t)

This is extremely important clinically.

D. Phase boundary detection

We must predict:

t_arterial_start
t_portal_start
t_late_start

This is basically **temporal segmentation**.

E. Tumor classification

Using:

- enhancement pattern
- lesion shape
- size
- washout timing

Predict:

Benign
Malignant
Subtype

Examples:

Tumor	Pattern
HCC	arterial hyperenhancement + late washout
Hemangioma	peripheral nodular enhancement
Metastasis	rim enhancement

---

2. The overall pipeline (high level)

Typical system:

Video
↓
Frame feature extraction
↓
Lesion detection
↓
Lesion tracking
↓
Temporal modeling
↓
Enhancement curve extraction
↓
Phase detection
↓
Tumor classification

But architectures can combine these steps.

---

3. Key challenges in CEUS

Important to understand before designing models.

1. Speckle noise

Ultrasound contains heavy noise.

2. Motion

Motion comes from:

- breathing
- probe movement
- tissue deformation

3. Very small lesions

Sometimes only **10–20 pixels**.

4. Long temporal dependency

2 minute video ≈ **3600 frames** (30 FPS)

Models cannot process all frames easily.

---

4. Major architecture families

We can categorize models into **7 main groups**:

1. Frame-based CNN pipelines
2. CNN + RNN temporal models
3. 3D CNN spatiotemporal models
4. Transformer-based video models
5. Detection + tracking pipelines
6. Segmentation-based pipelines
7. Self-supervised / foundation models

Now we go deep into each.

---

5. Approach 1 — Frame-based CNN + temporal postprocessing

Architecture

Video
↓
Frame sampling
↓
2D CNN detector
↓
Tracking algorithm
↓

Intensity curve
↓
Rule-based phase detection
↓
Classifier

Model choices

Detector:

- YOLOv8
- Faster R-CNN
- RetinaNet

Tracking:

- SORT
- DeepSORT

Why this approach works

Lesions usually remain **spatially localized**.

Once detected:

- track bounding box
- compute average intensity inside lesion

Then derive:

enhancement_curve(t)

From the curve:

- detect wash-in

- detect wash-out

Advantages

✔ very interpretable
✔ easy to implement
✔ low compute
✔ small dataset sufficient

Disadvantages

❌ temporal reasoning weak
❌ detection errors propagate
❌ cannot model dynamic enhancement patterns well

---

6. Approach 2 — CNN + RNN (classic medical video architecture)

Very popular in medical imaging.

Architecture

Frame
↓
CNN encoder
↓
Feature vector
↓
LSTM / GRU
↓
Temporal representation
↓
Phase prediction + tumor classification

CNN extracts **spatial features**.

RNN learns **temporal evolution**.

CNN options

- ResNet
- EfficientNet
- DenseNet

Temporal models

- LSTM
- GRU
- Bidirectional LSTM

Example pipeline

```
frame_t → CNN → f_t
[f_1, f_2, f_3 ... f_T]
      ↓
    LSTM
      ↓
Temporal embedding
      ↓
Phase classification
```

Why this works

Lesion enhancement follows **temporal patterns**.

RNN learns patterns like:

increase → peak → decrease

Which correspond to CEUS phases.

Advantages

✔ good temporal modeling
✔ relatively simple
✔ widely used in medical video

Disadvantages

❌ RNN struggles with very long sequences
❌ sequential computation (slow)
❌ limited global temporal attention

---

7. Approach 3 — 3D CNN models

These process **spatiotemporal volumes directly**.

Example architecture:

Input: 16-frame clip
↓
3D convolution
↓
spatiotemporal features
↓
classifier

Example models

- C3D
- I3D
- SlowFast Network

Why they work

3D CNN kernels learn:

spatial + motion patterns

Example kernel:

3×3×3

captures:

- x dimension
- y dimension
- time

Advantages

✔ joint spatial + temporal modeling
✔ strong performance in video tasks
✔ easier training than transformers

Disadvantages

❌ very heavy memory
❌ limited temporal range
❌ needs lots of data

---

8. Approach 4 — Video Transformers (state-of-the-art)

Modern research direction.

Architecture

Video frames
↓
Patch embedding
↓
Temporal + spatial attention
↓
Transformer encoder
↓
classification heads

Popular models

- TimeSformer
- Video Swin Transformer
- ViViT

Why transformers are powerful

They can model **long-range dependencies**.

Example:

Frame 20 → Frame 1500 relation

Important because CEUS phases occur **tens of seconds apart**.

Attention example

Transformer learns relationships like:

arterial peak → later washout

Advantages

✔ state-of-the-art performance
✔ global temporal modeling
✔ flexible architectures

Disadvantages

❌ extremely large compute
❌ needs big datasets
❌ complex training

---

9. Approach 5 — Detection + tracking + temporal model (very practical)

This is often the **best medical pipeline**.

Architecture:

Video
↓
Lesion detector
↓
Lesion tracker
↓
ROI crop
↓
Temporal network
↓
Phase + tumor classifier

Detector options:

- YOLOv8
- DETR

Why this works well

Instead of processing whole frame:

focus only on lesion ROI

This improves:

- signal-to-noise
- computational efficiency

Advantages

✔ efficient
✔ interpretable
✔ modular system

Disadvantages

❌ detection failure ruins pipeline
❌ multiple modules to train

---

10. Approach 6 — Segmentation based models

Instead of bounding boxes, detect **exact lesion mask**.

Models:

- U-Net
- U-Net++
- nnU-Net

Why segmentation is useful

You can compute **accurate enhancement curves**.

Example:

mean_intensity(mask, t)

This gives precise CEUS curve.

Advantages

✔ precise lesion measurement
✔ strong medical interpretability
✔ good for radiologists

Disadvantages

❌ annotation expensive
❌ segmentation harder than detection

---

11. Approach 7 — Self-supervised video representation learning

Important because **medical datasets are small**.

Techniques:

- contrastive learning
- masked video modeling

Models:

- VideoMAE

- SimCLR

Idea

Pretrain on **unlabeled ultrasound videos**.

Learn:

- motion
- anatomy
- texture

Then fine-tune for lesion detection.

Advantages

✔ less labeled data needed
✔ better generalization

Disadvantages

❌ more complex training

---

12. Most promising architecture (research perspective)

If I were designing a **state-of-the-art CEUS system**, I would do:

Hybrid architecture

Video

↓
YOLOv8 lesion detection
↓
DeepSORT tracking
↓
ROI cropping
↓
Video Swin Transformer
↓
Multi-task head
  ├ phase detection
  ├ enhancement curve prediction
  └ tumor classification

Multi-task learning improves performance.

---

13. Evaluation metrics

Important benchmarks:

Detection:

mAP
IoU

Segmentation:

Dice score
Jaccard index

Classification:

AUC
Accuracy
F1 score

Phase detection:

Temporal IoU
phase boundary error

---

14. Research ideas you could explore

Interesting directions:

1️⃣ Phase-aware networks

Model explicitly learns CEUS phases.

2️⃣ Physics-informed networks

Use contrast dynamics models.

3️⃣ Graph neural networks

Track lesion appearance changes.

4️⃣ Multimodal models

Combine:

- B-mode ultrasound
- CEUS video
- clinical metadata

---

15. My honest recommendation for your project

Start with:

Stage 1

YOLOv8 lesion detection
+
DeepSORT tracking

Then:

Stage 2

ROI sequence → CNN + BiLSTM

Then if dataset large:

Stage 3

Video Swin Transformer

Tab 2

I'll give you a **deep research-level overview** of CEUS liver lesion AI so you can actually design a **publishable system**. I'll cover:

1. How radiologists actually diagnose CEUS liver lesions
2. Latest research directions (2023–2025)
3. Public datasets you can use
4. A **complete research-grade architecture**
5. Open research problems (good for publishing papers)

---

1. How Radiologists Diagnose CEUS Liver Lesions

Understanding the **clinical reasoning** is extremely important because AI models should replicate this workflow.

CEUS basics

Contrast agents contain **microbubbles** injected into the bloodstream. These bubbles reflect ultrasound strongly.

CEUS captures **blood perfusion dynamics** in real time.

Radiologists observe how the lesion behaves after injection.

CEUS provides **dynamic perfusion information**, which helps distinguish benign from malignant lesions.

---

2. CEUS Phases Radiologists Use

There are **three major phases**.

1 Arterial Phase

Time:

10–30 seconds after injection

Characteristics:

- rapid enhancement
- blood enters lesion

Radiologists check:

- hyperenhancement
- rim enhancement
- peripheral nodular pattern

Important tumors:

Tumor	Pattern
Hepatocellular carcinoma (HCC)	arterial hyperenhancement
Hemangioma	peripheral nodular enhancement
Metastasis	rim enhancement

---

2 Portal Venous Phase

Time:

30–120 seconds

Radiologists check:

- whether contrast remains

- washout patterns

Example:

Malignant tumors often show **early washout**.

3️⃣ Late Phase

Time:

>120 seconds

Important observation:

washout strength

Malignant lesions typically show:

hypo-enhancement

---

3. Radiologists Use the Time-Intensity Curve

Radiologists sometimes compute:

TIC = Time-Intensity Curve

Example curve:

```
Intensity
  ^
  |        /\ peak
  |       /  \
  |      /    \
  |_____/      \____
  ----------------------> time
```

Important parameters:

Feature	Meaning
Time to peak	arterial flow speed
Wash-in slope	perfusion
Washout time	malignancy indicator
Peak intensity	vascular density

Machine learning systems often extract these features.

---

4. Real AI Systems for CEUS (Recent Papers)

Several modern studies exist.

CNN-LSTM CEUS model

A 2024 study used:

2D CNN + LSTM

Input:

CEUS video frames

Output:

benign vs malignant

Performance:

AUC ≈ 0.91

which outperformed several classical approaches.

Multi-phase CNN model

Another study trained **three parallel CNN networks** using images from different CEUS phases to classify lesions.

Dataset:

181 lesions

Classes:

- benign
- hepatocellular carcinoma
- non-HCC malignant

This showed that **multi-phase information is essential**.

Multimodal CEUS AI model (2025)

A newer model combined:

- CEUS videos
- biomarkers
- clinical data

The system classified **six liver lesion types** and achieved performance comparable to experienced radiologists.

---

5. Datasets for CEUS Liver Lesions

Public datasets are limited, but some exist.

---

1 B-mode + CEUS Liver Dataset

Source: The Cancer Imaging Archive

Size:

120 patients
272 GB

Contains:

- B-mode ultrasound
- CEUS videos
- clinical diagnosis

Collected during clinical trials.

This dataset is extremely useful for training.

2️⃣ SYSU-CEUS Dataset

Contains:

Lesion type	Cases
HCC	186
Hemangioma	109
FNH	58

Resolution:

768×576

Collected at Sun Yat-sen University hospital.

3️⃣ Private clinical datasets

Most CEUS research uses **hospital datasets**.

Typical size:

200–1000 patients

Because CEUS is a specialized procedure.

---

6. The Most Important CEUS Features

If you want a **good ML system**, extract these features.

Spatial features

From lesion region:

- texture
- shape
- margin
- size
- echogenicity

Temporal features

From CEUS dynamics:

- wash-in speed
- wash-out speed
- peak intensity
- enhancement heterogeneity

Radiomics features

Examples:

- GLCM texture
- wavelet features
- entropy
- local contrast

Radiomics are often used with ML classifiers.

---

7. A Full Research-Grade Architecture

This is a **strong architecture suitable for a paper**.

Stage 1 — Liver Segmentation

First isolate the liver region.

Model:

U-Net

Reason:

- removes background
- reduces false positives

Stage 2 — Lesion Detection

Use object detection.

Models:

- YOLOv8
- DETR

Output:

bounding box

Stage 3 — Lesion Tracking

Track lesion across frames.

Methods:

Kalman filter
DeepSORT
optical flow

Purpose:

Maintain lesion identity.

Stage 4 — ROI Sequence Extraction

Crop lesion region across time.

Result:

lesion video clip

Stage 5 — Temporal Modeling

This is the most important stage.

Best models:

Option A — CNN + LSTM

ResNet → LSTM → classifier

Option B — 3D CNN

3D ResNet

Option C — Transformer

Video Swin Transformer

Transformers are becoming **state-of-the-art**.

Stage 6 — Multi-Task Prediction

Instead of predicting only classification, predict multiple outputs.

Output heads:

phase classification
tumor classification
enhancement curve regression
lesion segmentation

Multi-task learning improves performance.

---

8. Advanced Architecture (Research Idea)

A very strong architecture could be:

Video
↓

Liver segmentation
↓
Lesion detection
↓
ROI extraction
↓
Video Transformer
↓
Temporal attention
↓
Multi-task prediction

Enhancements:

- contrast dynamics module
- physics-informed modeling
- self-supervised pretraining

---

9. New Research Direction (2024–2025)

Important new ideas.

1️⃣ Attention-guided CEUS models

Example:

TASL-Net

Uses:

- temporal attention
- spatial attention
- perfusion attention

to mimic how radiologists inspect ultrasound videos.

---

2 Optical Flow + Deep Learning

Used to detect **blood perfusion motion**.

Flow reveals:

contrast propagation

---

3 Multimodal AI

Combine:

- CEUS
- CT
- MRI
- lab results

---

4 Self-supervised ultrasound models

Pretrain on large ultrasound datasets.

---

10. Evaluation Metrics

Detection:

mAP
IoU

Segmentation:

Dice score

Temporal prediction:

phase accuracy
temporal IOU

Classification:

AUC
F1

---

11. Open Research Problems (Great for Publications)

These are still unsolved.

1️⃣ Motion-robust lesion tracking

Ultrasound motion is very complex.

2️⃣ Very small lesion detection

Some lesions <10 pixels.

3️⃣ Explainable AI for CEUS

Doctors need:

visual explanation

Example:

attention heatmaps

4️⃣ Real-time CEUS diagnosis

Goal:

diagnosis during ultrasound scan

---

12. If You Want a Strong Research Paper

I would propose this architecture:

YOLOv8 lesion detector
↓
DeepSORT tracker
↓
ROI extraction
↓
Video Swin Transformer
↓
Temporal contrast dynamics module
↓
Multi-task head

Outputs:

phase prediction
tumor classification
enhancement curve

This would be **a strong MICCAI / MedIA paper idea**.

Tab 3

I'll walk you through the **true state-of-the-art (SOTA) research landscape** for **CEUS liver lesion detection and classification**, including the **main papers, architectures, and research trends**. I'll structure it like a **research survey**, so you can quickly understand the evolution of the field and where the frontier is.

---

1. What the CEUS AI field actually focuses on

Almost all research in CEUS liver AI tries to solve **three core problems**:

1️⃣ Lesion detection / segmentation

Locate the lesion in ultrasound frames.

Typical outputs:

- bounding box
- segmentation mask
- lesion size

2️⃣ Enhancement pattern modeling

Analyze how contrast evolves over time.

This includes:

- arterial enhancement
- portal venous washout
- late phase intensity

These patterns determine tumor type.

3️⃣ Tumor classification

Common classes:

Tumor type	Category
Hepatocellular carcinoma	malignant
Metastasis	malignant
Cholangiocarcinoma	malignant
Hemangioma	benign
Focal nodular hyperplasia	benign

---

2. Early SOTA (2017–2020)

These were mostly **radiomics + classical ML**.

Typical pipeline:

CEUS video
↓
Lesion ROI extraction
↓
Time-intensity curve
↓
Feature extraction
↓
ML classifier

Common models:

- Random Forest
- SVM
- Artificial Neural Networks

Example result:

- ANN models achieved around **87% accuracy** for classifying five tumor types from CEUS images.

Limitation

These systems depended heavily on **handcrafted features**, such as:

- washout time
- peak intensity
- TIC slope

This limited performance.

---

3. Deep Learning SOTA (2020–2022)

Researchers started using **deep CNNs**.

Typical architecture:

CEUS frames
↓
CNN encoder
↓
feature vector
↓
classifier

One system trained deep neural networks to classify **five liver tumor types** using CEUS and achieved **~88% accuracy**.

Common architectures

- ResNet
- DenseNet
- VGG

But these models treated frames **independently**, ignoring temporal dynamics.

---

4. Temporal Deep Learning (2022–2024)

This is where the field changed dramatically.

Researchers realized that **contrast dynamics over time** are crucial.

So models started using:

CNN + temporal models

Example: CNN + LSTM model (2024)

Architecture:

CEUS video frames
↓
2D CNN
↓
LSTM temporal encoder
↓
classifier

Dataset:

420 liver lesions

Performance:

AUC = 0.91

This outperformed:

- 3D CNN
- classical ML using TIC features.

Why CNN-LSTM works well

CNN learns:

spatial features

LSTM learns:

contrast evolution

which mimics how radiologists interpret CEUS.

---

5. Multi-Phase Deep Networks

Another major idea is **phase-aware modeling**.

CEUS contains multiple phases:

- arterial
- portal
- late
- Kupffer

Researchers train networks using **separate phase inputs**.

Example: Multi-phase ResNet model

Architecture:

9 CEUS phase images
↓
3 parallel ResNet50 networks
↓
fusion layer
↓
classifier

This system achieved:

- **93% sensitivity** for detecting malignant lesions.

Key idea

Each phase provides **different diagnostic information**.

Example:

Phase	Important feature
arterial	hyperenhancement
portal	washout
late	hypo-enhancement

---

6. Multicenter AI Systems (2025 SOTA)

The most advanced research now uses **large multicenter datasets**.

One 2025 study developed a CEUS-based AI system using **3,725 liver lesions from 52 medical centers**.

The model classified **six lesion types**:

- hepatocellular carcinoma

- metastasis
- cholangiocarcinoma
- hemangioma
- abscess
- others

Architecture

The system combined **three modules**:

Module-Disease
Module-Biomarker
Module-Clinical

These were fused into one model.

Performance

Accuracy:

0.85–0.86

which was:

- better than **junior radiologists**
- comparable to **senior radiologists**.

---

7. Attention-based Video Models (New SOTA)

Recent work uses **transformer + attention models**.

Example architecture:

Tri-Attention Selective Learning Network:

- temporal attention
- spatial attention
- perfusion attention

The model mimics how sonographers read CEUS videos.

Key innovations

1️⃣ **Time-Intensity-Curve guided attention**

Focus on frames where contrast changes.

2️⃣ **Earliest enhancement detector**

Find the first region that enhances.

3️⃣ **Bimodal encoding**

Uses:

- B-mode ultrasound
- CEUS video

---

8. Optical-Flow CEUS Models

Another emerging idea is using **optical flow**.

Contrast microbubbles move through vessels.

Optical flow captures:

blood perfusion motion

Recent systems combine:

optical flow + deep learning

to model contrast propagation.

---

9. Hybrid Radiomics + Deep Learning

Some models combine:

deep CNN features
+
radiomics features

Radiomics features include:

- texture
- entropy
- wavelets
- shape descriptors

These sometimes improve performance.

---

10. Real-Time CEUS AI

A major goal now is **real-time diagnosis during ultrasound scanning**.

AI can assist with:

- automatic lesion detection
- automatic phase recognition
- real-time tumor classification

This reduces dependence on expert radiologists.

AI-augmented CEUS systems are expected to improve diagnostic consistency and reduce operator variability.

---

11. Current SOTA Architecture (2024–2025)

Most cutting-edge systems look like this:

CEUS video
↓
Liver segmentation
↓
Lesion detection
↓
Lesion tracking
↓
ROI video extraction
↓
Video transformer
↓
multi-task prediction

Outputs:

- lesion classification
- phase detection
- enhancement curve
- LI-RADS category

---

12. Emerging Research Trends (2025–2026)

These are **active research directions**.

1️⃣ Video transformers

Examples:

- TimeSformer
- Video Swin Transformer
- ViViT

Advantages:

- long temporal modeling
- global attention

2️⃣ Self-supervised ultrasound learning

Labeling CEUS is expensive.

Researchers pretrain models on:

unlabeled ultrasound videos

3️⃣ Multimodal medical AI

Combine:

- CEUS

- CT
- MRI
- lab biomarkers

This is extremely promising.

4️⃣ Explainable CEUS AI

Doctors need explanations.

Methods include:

- Grad-CAM
- attention maps
- perfusion maps

---

13. Performance Benchmarks

Typical results across papers:

Model type	AUC
radiomics ML	0.75–0.80
CNN	0.85
CNN + LSTM	~0.91
multimodal deep models	0.90–0.94

---

14. Biggest Open Problems (still unsolved)

These are **excellent research opportunities**.

---

1 Motion-robust lesion tracking

Ultrasound motion is very complex.

---

2 Small lesion detection

Lesions can be **<5 mm**.

---

3 Phase boundary detection

Few papers solve:

automatic phase segmentation

---

4 Limited datasets

Most datasets have:

<500 patients

---

15. If you want to build a truly SOTA research model

A strong architecture today would be:

Video
↓
YOLO lesion detector
↓
lesion tracker

↓
ROI sequence
↓
Video Swin Transformer
↓
Temporal attention
↓
multi-task outputs

Outputs:

- phase prediction
- lesion segmentation
- tumor classification
- enhancement curve